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Splicing Image Forgery Detection by Deploying Deep Learning Model

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***Abstract*— The use of digital images has become more important in numerous industries, including journalism, medical diagnosis, law enforcement, and forensics. Images could be easily modified due to the availability of photo-altering tools and software, which can distort the original information. The availability of low-cost and simple image-editing software has also drastically decreased the time and money needed to engage in image tampering. This destroys the purity of the image, making it vulnerable to misuse by anyone. Copy-move, splicing, and enhancement falsification are prominent methods used to create fake images. Using a combination of two or more images, a new image is created called splicing, which may be shared across many online channels and used to spread information and impact target audiences. The discovery could have beneficial as well as negative results. Consequently, it is important to provide a method that can identify the splicing forgery in an image. The study used a Deep Learning (DL) model named MobileNet to distinguish between authentic and manipulated images. Both the training and testing of the DL model make use of the CASIA data. Resizing and enhancing methods are used to process the collected data. Accuracy, F1 score, and recall are employed to test the reliability of the created DL model.**

Keywords*— *Image forgery, Splicing, Deep Learning, Accuracy, MobileNet.

I. Introduction

Digital images surround us at all times. Magazines, newspapers, the education and insurance industries, the medical profession, the entertainment industry, matrimonial websites, and many other industries rely significantly on digital images. Because image editing software has grown so intuitive and simple to use, anyone can now modify images. In several ways, digital images could be altered or created from scratch. Most instances of image forgery include some form of image splicing or copy-move forgery. Image splicing forgery involves combining various images or sections of images to generate a single falsified image. The copy-move forgery, on the other hand, involves copying an image's components in multiple locations within the same image. Image falsification has become increasingly common in recent years. Campaigning on social media platforms is increasingly standard practice in competitions all around the world. On the plus side, digital images are routinely used to boost voter awareness of the election. Meanwhile, fake images and images with incorrect information were widely shared online to influence public perceptions. Research [1] done during the latest Brazilian presidential elections found that about 13.1 percent of messages sent via WhatsApp were false. Recently, many fake images with misleading data on the COVID-19 outbreak spread rapidly on social networks [2]. As a result, it is critical to build images for authenticating images to re-establish trust in digital photography.

Several authors have presented several approaches to verify the images in recent years. Totally two varieties of forgery detection: active and passive [3,4]. Understanding the image data is required by any active forgery detection technique. The opposite is true with passive forgery detection methods, which expose the fake by retrieving particular clues based on modifications in the image's fundamental properties and need no previous understanding of the source images. Unfortunately, the bulk of stolen images is obtained from numerous social networking sites. The veracity of such manufactured images becomes extremely difficult to detect. As a result, active forgery detection methods are less successful than they were previously. Passive forgery detection methods are ideal for detecting a fake in digital images.

Using DL, the publication [5] presents an image for detecting splicing forgery in images. There are three phases to the suggested technique: 1) Image pre-processing activities include converting images to RGB and changing image sizes. 2) Utilizing the built Convolutional Neural Network (CNN) AlexNet network, to extract the overall discriminative feature by a cleaned-up image. 3) The third stage involves employing the obtained feature representation to create a Canonical Correlation Analysis (CCA) predictor for true/false labelling. When tested on the CASIA splicing forgery detection dataset utilizing the AlexNet network based on characteristics with a CCA predictor, the suggested approach shows promising accuracy. When

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compared to the current state of affairs, the recommended strategy is superior. In the paper [6], a SE-Network (Squeeze & Excitation network) is developed to locate the splicing area. The suggested framework employs a convolutional encoder-decoder architecture with two streams: label mask and edge-guided. Edge-guided data is sent to the labelled masking channel to enhance feature separation between edited and parent regions. There are three main accomplishments of this research. First, determine the location of spliced DNA. Second, incorporates localization dependability as a feature. Finally, fine-tune the fusion parameters to maximize feature expression. Extensive studies on 2 synthetic and 4 standard splicing databases demonstrate that the suggested SE-Network is superior to the existing splicing localization methods. The authors of [7] suggest an end-to-end framework that can not only detect and localize spliced regions but also differentiate between regions originating from distinct donor images. To identify spliced portions from multiple sources apart, they consider both spliced portions and background images and employ a localized approximation of the principal quantization matrix. To accomplish this, they first use a clustering algorithm to aggregate input images according to the anticipated quantization matrix, and then they use a morphological restoration to fine-tune the outcome. In addition to being usable across aligned and non-aligned double JPEG compression, as well as in cases where the latter compression is either greater or less than the previous, the suggested method's flexibility is another key advantage. Extensive experiments confirmed that the proposed technique outperformed baseline procedures in identical conditions.

The goal of this research [8] was to develop a reliable forensic analyser for identifying instances of image splicing forgery. Inside the framework of the hypothesis's methodology, the analyser is built on a modified noise technique of a JPEG image for identifying image forgery. The developed detector's prediction accuracy can be optimized by altering the false alarm rate. However, the threshold for making a decision is flexible and independent of any one aspect of an image. Furthermore, the statistical performance of the detector may be shown analytically using strong theoretical explanations. Numerical statistics have validated the detector's efficiency and advantage over existing comparator detectors. The article [9] proposes edge-enhanced Transformers (ET) for locating vulnerable locations. A two-branch edge-aware transformer is created to include spliced edge indications into a fraud localization system. This produces forgery and edge features, that are subsequently utilized to gather abundant evidence of manipulation. They also create a feature enhancement mechanism that calls an emphasis on forgery's edge region artifacts via introducing weighting factors to the resultant matrix in the spatial domain, thus amplifying the signal and reducing the noise. Experiment results using benchmark data show that the proposed method surpasses state-of-the-art strategies for detecting tampered areas at the pixel and edge levels. In the paper [10], the author looks into the problem of spotting splicing operations in amplitude Synthetic Aperture Radar (SAR) images. The purpose of this work is to identify illegally modified image pixels in amplitude SAR data while keeping in mind that the attacker may have made alterations to conceal their image. The method employs a CNN to highlight anomalies in the processed traces of the input under consideration. The fingerprint is then analysed to build a binary tampering mask of the spliced pixel region. The outcome indicates that the suggested SAR-specific approach surpasses the most cutting-edge forensic tools built for natural images.

The organization of paper is as follows: Part I describes the digital image forgery detection techniques carried out in recent years. Part II details the methodology of the suggested framework to identify the forgery detection from the splicing images. Part III demonstrates the result attained by the designed model in training as well as testing phase. Part IV concludes the research by highlighting the designed model metrics.

II. Methods and Materials

The DL-based image spicing detection using a standard dataset is detailed in this section. The work carried out to accomplish the task is given in figure 1. The data is collected from the standard benchmark database. After data collection, the data will be passed through processing techniques like data size conversion and color enhancement. Next to data processing, the processed data is split into two divisions like train and test. The ratio considered for train and test division is 8:2. Then, the DL model is constructed. The developed DL model was first trained using 80% of the collected data and tested by utilizing the remaining 20% of the data. The metrics score like accuracy, F1-score, and recall is employed to evaluate the performance of the suggested DL model in both the training and testing model.

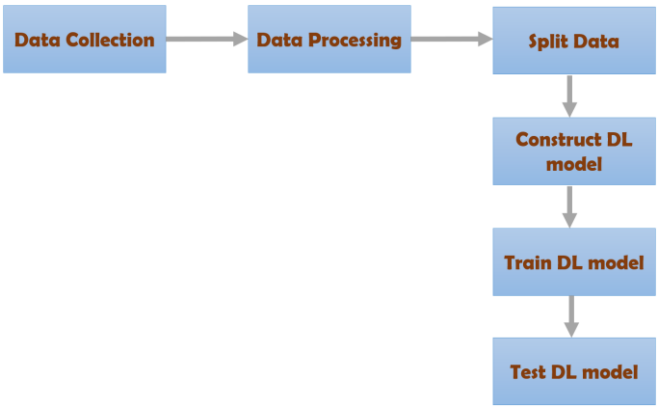

Fig. 1. Splicing Image Detection using DL

A. Data Collection

There are two versions of the CASIA dataset, numbered 1.0 and 2.0, respectively. For this research, we utilize CASIA 1.0 [11]. Version 1.0 of CASIA is dedicated to splicing detection, and it contains 800 authentic and 921 fake (spliced) 384x256 pixel JPEG color images. The 384 x 256 image JPEG color images included in this collection are split evenly between 800 real and 921 fakes (spliced). There are eight distinct groups in this particular dataset (Nature, buildings, textures, scenes, characters, plants, and other flora and fauna). The sample CASIA data is given in figure 2.

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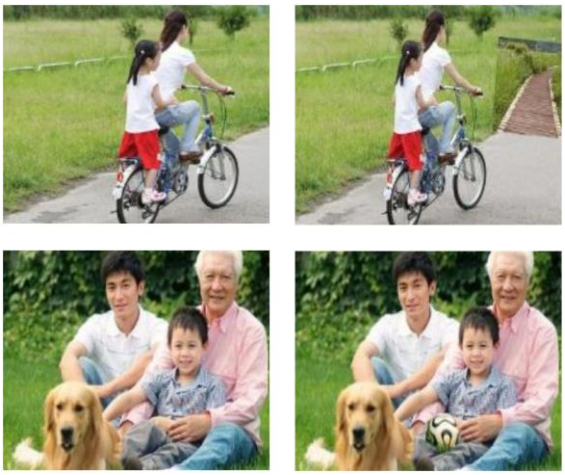

Fig. 2. Sample CASIA V1.0 Dataset

B. Data Processing

As part of the pre-processing step, a variety of processes are done to images, and distortions are corrected, which improves the image of the image data. It also has the potential to be used to improve specific yet important components of an image for subsequent processing. Simple operations on images, such as scaling, can be classified as "image processing" [12]. For example, to use a specific image dataset with a DL model, all images inside the dataset must be the same size. The Scaling is defined by a consistent mapping between the image and image pixels. The most frequent scaling approach is an interpolation of the pixels in the original image. These interpolation scaling approaches might cause artefacts such as fake blocks and aliasing. If the aspect ratios of the input and output images differ significantly, the scaling image will exhibit visible distortion. During pre-processing, the noise and distortions in the input images are eliminated, improving image quality. The relative number of dark and light pixels in an image determines image contrast. A low-contrast image has little discernible contrast between its bright and dark sections [13]. There are two types of ways for improving a person's image. Methods for dealing with space. The histogram equalization method is frequently used to improve images. Histogram equalization can be used to improve global contrast when the image's relevant information is characterized by relatively low contrast values. Low pass filtering can be used to remove the image's high-frequency features for a more blurred effect.

C. DL model

The MobileNet is chosen as a DL model to predict the image forgery. CNN has found widespread application in image classification and segmentation in computer vision [14]. Modern CNNs, on the other hand, are becoming more involved and sophisticated to increase their accuracy. In MobileNet [15], both the count of parameters and the count of multiplication and addition (multi-adds) operations were decreased. MobileNets are built on depthwise separable convolution layers (DSCLs). Each DSCL comprises a depth and a pointwise convolution. The normal convolution approach multiplies each image pixel by the number of filter channels, then adds the pixels handled by the filter as it passes across the entire image. The depthwise separable final Convolution Layer (CL) is used for channel-based image property learning. With depth-separable convolution, huge kernels can be broken into several tiny kernels that can carry out the exact work in fewer iterations. Some of these methods include the simultaneous processing of two procedures, like depthwise convolution followed by pointwise convolution.

Figure 3 depicts the internal structure of MobileNet. The 1st CL has a stride of 2. Following that, the depthwise layer (DL) and pointwise layer (PL) swap places. As data moves through the network model, its dimensions (width and height) shrink, hence the stride of DL is set to 1 and 2, correspondingly. The addition of the PL increases the count of channels in the information to four. Following the CL is an activation function based on the ReLU tensor. This process is repeated until the original 224x224-pixel image is shrunk to 7x7 pixels with 1024 channels. Last but not least, an average pooling operation was performed, yielding a 1 × 1 × 1024-pixel image. Then reduce the network size and speed by adjusting the following hyperparameters.

1. The width multiplier can be used to alter the channel depth by entering a value between 0 and 1.
2. The resolution multiplier, denoted by, modifies the size of an input image by a factor ranging from 0 to 1.

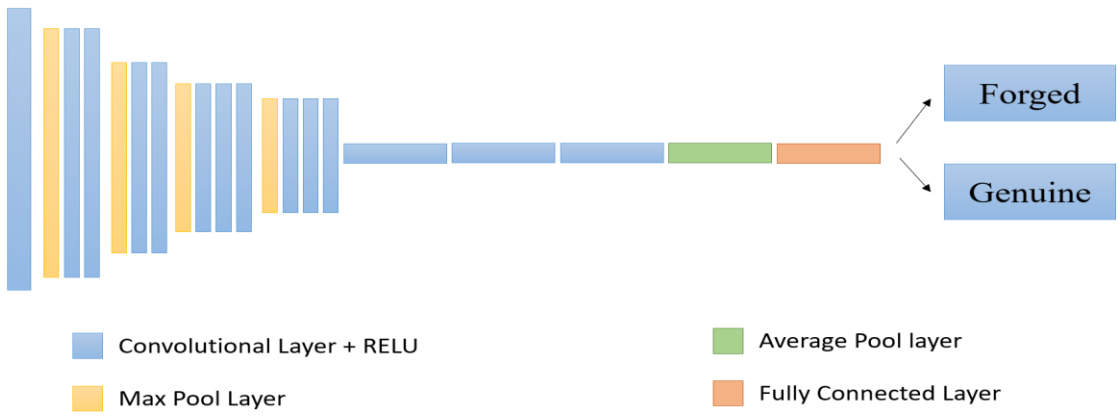

Fig. 3. MobileNet Architecture

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III. Experimental Result

The CASIA V1.0 data is gathered to design a DL-based image splicing detection system. The collected data is not good to directly apply to the DL model. So, the data must take to some processing techniques called resizing and enhancing. The processed data is given to the DL model for training purposes. To evaluate the training phase, the metrics like accuracy, F1-score, and recall are employed. A total of 80 epochs is employed to train the DL model for identifying the splice images. The sample of metrics score attained in the training phase for the first 10 epochs is detailed in table 1.

TABLE 1: DL Model Evaluation for Splice Image Detection

Epoch	F1 Score	Accuracy	Recall
0	0.200479	0.341898	0.329642
1	0.433931	0.307594	0.456227
2	0.559049	0.329657	0.425686
3	0.407383	0.3992	0.441641
4	0.497467	0.408379	0.450413
5	0.475148	0.440113	0.501131
6	0.551245	0.473335	0.514517
7	0.538045	0.466125	0.531037
8	0.559736	0.47123	0.553519
9	0.681787	0.440285	0.579688
10	0.614478	0.552713	0.588992

Next, figure 4 to 6 is employed to depict the metrics score attained by the DL model concerning each epoch. In all the figures the epochs are taken on the x-axis and the metrics score will be on the y-axis. The orange plot given in figure 4 depicts the F1-score and the plot shows that the F1-Score will be more than 0.8 after the epoch of 44. Figure 5's green plot displays accuracy, and it's clear that after the epoch of 18, accuracy will be higher than 0.8. Figure 6's blue plot indicates recall, and it's easy to see that after the epoch of 30 the recall will be well above 0.8.

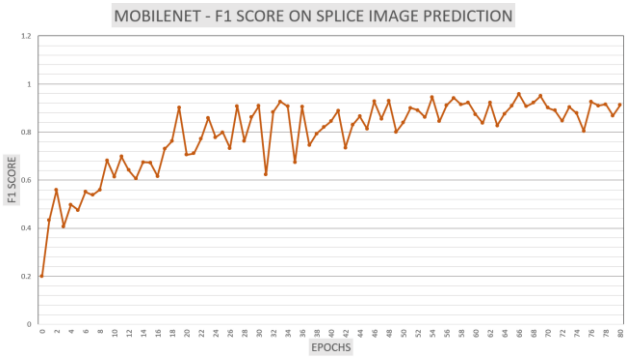

Fig. 4. F1 Score Plot of MobileNet

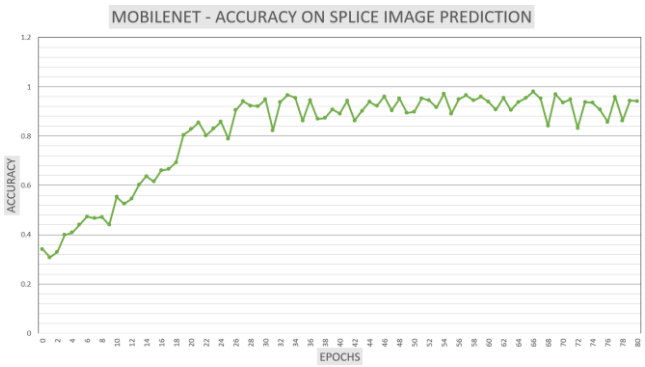

Fig. 5. Accuracy Plot of MobileNet

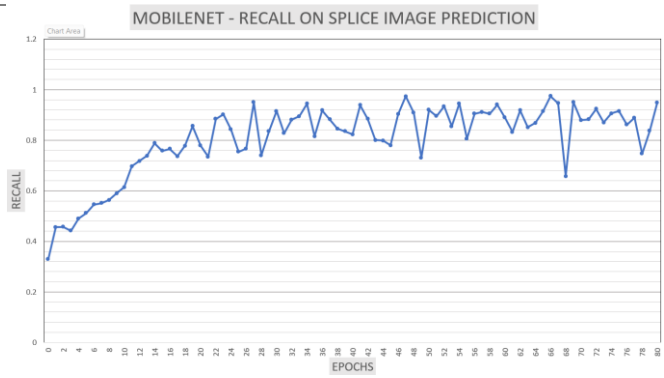

Fig. 6. Recall Plot of MobileNet

All figures 4, 5, and 6 give the hope that the DL model is eligible for identifying the image forgery. So, the architecture of the MobileNet is finalized, and test the model. The outcome of the MobileNet is given in figure 7. The accuracy of the model will be 0.9553, F1-score be 0.96271 and the recall score is 0.9514. After testing the model with overall data. The model is deployed to predict each image separately. The MobileNet image splicing prediction is given in figure 8. The two forged images are uploaded and ask the model to predict. Figure 8 shows that the actual and the model prediction outcome are the same.

```
45/45 [==============================] - 0s 375ms/step
– f1 score: 0.96271 - accuracy: 0.9553 - recall: 0.9514
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Fig. 7. Test Performance of MobileNet

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Actual: Forged
Predicted: Forged

Actual: Forged
Predicted: Forged

Fig. 8. MobileNet Prediction Outcome

IV. CONCLUSION

The development of accessible multimedia technologies has made image alteration a simple operation. Multiple modifications are possible to the images. One technique for manipulating images involves splicing together multiple images to create a new, unified one. Such images have the potential to trick people. This work makes an effort to address this problem by employing DL technology to detect fake images automatically. The CASIA V1.0 database served as the starting point for our investigation. Methods such as resizing and enhancement are applied to the collected image. Specifically, the data is evenly divided into 80/20 parts. The DL model is trained using 80% of images from the original data and evaluated using accuracy, f1, and recall. The remaining 20% of the data is then used to test the trained model. The model achieves an f1 score of 0.96271, an accuracy of 0.9553, and a recall score of 0.9514. The results from the DL model indicate that the created model will perform well in identifying forgeries in images.

REFERENCES

[1]. Machado C, Kira B, and Howard P N, “A Study of Misinformation in WhatsApp groups with a focus on the Brazilian Presidential Elections”, WWW ’19: Companion Proceedings of The 2019 World Wide Web Conference, pp 1013–9, 2019

[2]. Jessica McDonald, “Social Media Posts Spread Bogus Coronavirus Conspiracy Theory”, 2020

[3]. Y. Rao and J. Ni, "A deep learning approach to detection of splicing and copy-move forgeries in images," 2016 IEEE International Workshop on Information Forensics and Security (WIFS), 2016, pp. 1-6, doi: 10.1109/WIFS.2016.7823911.

[4]. S. D. Thepade, S. Bhandari, C. Bagde, R. Chaware and K. Lodha, "Image Forgery Detection using Machine Learning with Fusion of Global and Local Thepade's SBTC Features," 2021 International Conference on Disruptive Technologies for Multi-Disciplinary Research and Applications (CENTCON), 2021, pp. 234-238, doi: 10.1109/CENTCON52345.2021.9688094.

[5]. I. T. Ahmed, B. T. Hammad and N. Jamil, "Effective Deep Features for Image Splicing Detection," 2021 IEEE 11th International Conference on System Engineering and Technology (ICSET), 2021, pp. 189-193, doi: 10.1109/ICSET53708.2021.9612569.

[6]. Y. Zhang, G. Zhu, L. Wu, S. Kwong, H. Zhang and Y. Zhou, "Multi-Task SE-Network for Image Splicing Localization," in IEEE Transactions on Circuits and Systems for Video Technology, vol. 32, no. 7, pp. 4828-4840, July 2022, doi: 10.1109/TCSVT.2021.3123829.

[7]. Y. Niu, B. Tondi, Y. Zhao, R. Ni and M. Barni, "Image Splicing Detection, Localization and Attribution via JPEG Primary Quantization Matrix Estimation and Clustering," in IEEE Transactions on Information Forensics and Security, vol. 16, pp. 5397-5412, 2021, doi: 10.1109/TIFS.2021.3129654.

[8]. Y. Chen, F. Retraint and T. Qiao, "Detecting Spliced Image Based on Simplified Statistical Model," 2022 14th International Conference on Computer Research and Development (ICCRD), 2022, pp. 220-224, doi: 10.1109/ICCRD54409.2022.9730342.

[9]. Y. Sun, R. Ni and Y. Zhao, "ET: Edge-Enhanced Transformer for Image Splicing Detection," in IEEE Signal Processing Letters, vol. 29, pp. 1232-1236, 2022, doi: 10.1109/LSP.2022.3172617.

[10]. E. D. Cannas, N. Bonettini, S. Mandelli, P. Bestagini and S. Tubaro, "Amplitude SAR Imagery Splicing Localization," in IEEE Access, vol. 10, pp. 33882-33899, 2022, doi: 10.1109/ACCESS.2022.3161836.

[11]. CASIA Tampered Image Detection Evaluation Database (CASIA TIDE v1.0)” Available at: http://forensics.idealtest.org:8080/index_v1.html.

[12]. K. K. Pal and K. S. Sudeep, "Preprocessing for image classification by convolutional neural networks," 2016 IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT), 2016, pp. 1778-1781, doi: 10.1109/RTEICT.2016.7808140.

[13]. S.anitha , Dr.V.Radha, “Comparison of Image Preprocessing Techniques for Textile Texture Images ” International Journal of Engineering Science and Technology, vol. 2, no. 12, pp. 7619-7625, 2010

[14]. Kadam KD, Ahirrao S, Kotecha K. “Efficient Approach towards Detection and Identification of Copy Move and Image Splicing Forgeries Using Mask R-CNN with MobileNet V1”, Comput Intell Neurosci, 2022, doi: 10.1155/2022/6845326. PMID: 35035463; PMCID: PMC8754624.

[15]. A. G. Howard, M. Zhu, B. Chen et al., “MobileNets: efficient convolutional neural networks for mobile vision applications,” 2017, https://arxiv.org/abs/1704.04861.

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